

Current Numerical Results for the Bermudan Put Experiments

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Purpose of this note

This note collects the numerical results currently available from the standalone scripts. The idea is to have one short working document that can be shared and discussed easily. At this stage, the main point is to show where the benchmark stands, how the generalized double mean-reverting (gDMR) setup is behaving, and what changed after the recent tuning.

1. Paper benchmark: Heston Bermudan put

The `paper_setup` folder is kept as a benchmark against the Heston Bermudan put case reported by Farahany, Jackson, and Jaimungal. In this case we do have a published finite-difference reference value, so the comparison is quite direct.

Table 1: Benchmark comparison for the paper-style Heston setup.

Estimator	Published value	Current run	Run – FD	Run – published
LSMC direct	1.4494	1.458896	0.008196	+0.009496
LSMC low	1.4487	1.452544	0.001844	+0.003844
Hybrid direct	1.4530	1.452988	0.002288	-0.000012
Hybrid low	1.4529	1.445493	0.005207	-0.007407

The finite-difference reference for this benchmark is 1.4507. The most encouraging point here is that the hybrid direct estimator is essentially on top of the published hybrid direct value, while the other estimators remain in the same numerical range as the paper.

2. Current gDMR baseline: $\delta_1 = \delta_2 = 0.50$

For the gDMR case there is no published finite-difference benchmark available in the project, so the current practical reference is the large LSMC direct run. The default baseline uses the following setup.

The main result here is that the hybrid direct estimator is now within 1% of the LSMC direct reference. That is a clear improvement over the earlier hybrid runs and is, in my view, the strongest current numerical message from the gDMR experiments.

This progression suggests that the most important improvements were not cosmetic. The gains came mainly from three choices: a stronger Monte Carlo reference, truncation of the volatility regression domain, and a tighter asset grid around the strike.

Table 2: Main numerical settings for the current gDMR baseline run.

Quantity	Value
S_0	100
K	100
T	1
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
δ_1	0.50
δ_2	0.50
LSMC paths	1,000,000
Exercise dates	100
Euler steps	600
Hybrid paths N	30,000
Hybrid low paths N_{low}	30,000
Asset grid points N_S	181
Hermite nodes	64

Table 3: Final baseline results for the current gDMR setup.

Estimator	Price	Standard error	Relative error
LSMC direct	6.384535	0.007943	reference
LSMC low	6.387249	0.007939	0.04% vs LSMC direct
Hybrid direct	6.421758	0.000652	0.58% vs LSMC direct
Hybrid low	6.265070	0.045711	1.87% vs LSMC direct

3. Additional gDMR experiment: $\delta_1 = \delta_2 = 0.75$

I also ran a second gDMR experiment with $\delta_1 = \delta_2 = 0.75$, keeping the same tuned numerical configuration as in the current baseline. This gives a simple way to see how the price level changes when both elasticity parameters are increased.

The 1% direct-error target is still met in this equal-delta case. So, at least for the current parameter set, the hybrid direct estimator remains stable after moving from $\delta = 0.50$ to $\delta = 0.75$.

Numerically, all four estimators move upward when the model is changed from $\delta_1 = \delta_2 = 0.50$ to $\delta_1 = \delta_2 = 0.75$. At the same time, the agreement between the direct estimators remains quite good.

4. Main points to discuss

- The paper benchmark looks healthy: the standalone implementation reproduces the published Heston numbers reasonably well, especially for the hybrid direct estimator.
- In the gDMR case, the strongest current result is that the tuned hybrid direct estimator now sits within 0.58% of the LSMC direct reference.

- The low estimator is still visibly weaker than the direct estimator in the gDMR experiments, so this remains a natural place for further improvement.
- The equal-delta experiment with $\delta_1 = \delta_2 = 0.75$ raises the price level, but it does not break the current numerical behavior of the direct hybrid method.

5. Short conclusion

At the moment, I think the main story is quite clear. The benchmark case gives confidence that the implementation is behaving sensibly, and the tuned gDMR setup now produces a hybrid direct value that is close to the Monte Carlo reference. That feels like a solid basis for discussion with the supervisor, even if the low estimator still needs more work.

Table 4: Tuning path for the hybrid method in the gDMR baseline.

Run	Key settings	Hybrid direct	Hybrid low	Direct rel. error vs LSMC direct
Tier 1	10,000 paths, 121 grid points, 48 nodes, wide asset grid	7.335311	6.621170	14.89%
Tier 2	20,000 paths, 161 grid points, 48 nodes, wide asset grid	6.987310	6.511937	9.44%
Tier 3	40,000 paths, 201 grid points, 64 nodes, wide asset grid	6.804899	6.424076	6.58%
Truncated Tier 2	20,000 paths, 161 grid points, 48 nodes, truncation added, wide asset grid	6.756928	6.281426	5.84%
Previous default	20,000 paths, 161 grid points, 48 nodes, truncation and	6.486941	6.279662	1.60%

Table 5: Results for the equal-delta case $\delta_1 = \delta_2 = 0.75$.

Estimator	Price	Standard error	Relative error
LSMC direct	6.653560	0.007822	reference
LSMC low	6.655062	0.007819	0.02% vs LSMC direct
Hybrid direct	6.696152	0.000375	0.64% vs LSMC direct
Hybrid low	6.545257	0.045080	1.63% vs LSMC direct

Table 6: Comparison of the two gDMR cases currently tested.

Estimator	$\delta = 0.50$	$\delta = 0.75$	Change
LSMC direct	6.384535	6.653560	+0.269025
LSMC low	6.387249	6.655062	+0.267813
Hybrid direct	6.421758	6.696152	+0.274394
Hybrid low	6.265070	6.545257	+0.280187